

HRISHABH MAHAJU

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SUMMARY

Data Science M.S. candidate (UNH, May 2026) with 3+ years of production software engineering experience. Delivered measurable improvements in high-traffic systems across full-stack development, QA automation, cloud data pipelines, and AI agent development. Targeting Data Engineer, ML Engineer, or AI Developer roles.

EDUCATION

M.S. Data Science - University of New Haven, CT

Aug 2024 – May 2026

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Distributed & Scalable Data Engineering (AWS), Power BI

B.E. Computer Engineering - Pokhara University, Nepal

Sept 2016 – July 2022

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

ML / AI: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Keras, NLTK, OpenCV, Hugging Face, LangChain, RAG

Web / Backend: Spring Boot, Django, Flask, ReactJS, NextJS, Angular, NodeJS, REST APIs

Cloud & Data: AWS (S3, Kinesis, Lambda, Glue, Athena), Docker, Redis, MongoDB, MySQL, Firebase

Tools: Power BI, Git/GitHub, Selenium, JMeter, Postman, Burp Suite, JIRA, Scrum/Agile

WORK EXPERIENCE

AI Engineer Intern · Webstars LLC / AI Trusted Advisors · Remote

Oct 2025 – Present

Technologies: Python, Voice AI, AI Agents, MCP, LangChain, ReactJS, NodeJS | Methodology: Agile

- Designing and developing custom Voice AI agents and MCP platform solutions to support client acquisition, internal automation, and partner integrations for a U.S.-focused AI advisory startup.
- Building backend infrastructure for agent/partner access portals, including usage tracking, billing, and platform reliability; responsible for weekly backups and outage response.
- Developing AI-powered cold outreach automation with A/B testing pipelines targeting 50,000–100,000+ emails/month, and building social media AI agents across LinkedIn, Reddit, Instagram, and TikTok.

Software Engineer / QA Automation Engineer · YCO Pvt. Ltd. (Nepal Stock Exchange) · Kathmandu, Nepal

Oct 2021 – Jun 2024

Stack: Java, Spring Boot, Angular, Python, Selenium, JMeter, Redis, Docker, MSSQL, BIRT/Jasper | Methodology: Scrum

- Improved software performance by 50% through full automation testing (functional, regression, performance) across a microservices-based Fund Management and Trade Management System for Nepal's national stock exchange.
- Implemented High Availability (HA) multi-server architecture for nepalstock.com using Spring Boot, boosting system reliability and driving a 30% improvement in user engagement under peak-traffic conditions.
- Engineered Redis caching layer and optimized SQL stored procedures for high-throughput real-time data feeds, measurably reducing database load and API response times at national scale.
- Acted as Product Owner: owned QA backlog, prioritized features across 3 cross-functional teams, and conducted RCA investigations that cut recurring production incidents by ~40%.
- Containerized services with Docker; developed custom BIRT/Jasper reports replacing manual stakeholder reporting workflows; contributed to RBAC security refactoring for the exchange website.

Freelance Full-Stack Developer · Motorable Local Roads Bridge Program (MLRBP) · Nepal

Apr – May 2024

Stack: ReactJS, NextJS, React-Leaflet, GeoJSON, MariaDB, Docker | Live: webmap.lrbpnepal.org

- Built end-to-end geospatial web app visualizing 1,000+ bridge projects nationwide using React-Leaflet and GeoJSON; designed MariaDB schemas for large-scale real-time data and containerized full stack with Docker.

PROJECTS

Real-Time Traffic Analytics Pipeline (AWS Kinesis · S3 · Glue · Athena · Lambda) — Cloud-native pipeline ingesting and analyzing simulated web traffic logs with end-to-end operational intelligence.

DocuMate: AI PDF Assistant (Python · LangChain · FAISS · RAG · Streamlit) — Conversational LLM assistant using RAG and vector-similarity retrieval to answer contextual questions from PDFs.

Alzheimer's BRFS Analysis (Python · Pandas · K-Means · Matplotlib) — Trend analysis, confidence intervals, and clustering on CDC survey data to surface demographic health risk patterns.

E-Health Care Chatbot (Python · Decision Tree · Flask) — Symptom-based disease prediction chatbot with mental health consultation flow using a Decision Tree classifier.

CERTIFICATIONS

- MLOps Essentials: Model Deployment & Monitoring (LinkedIn Learning, Apr 2025) & MLOps Essentials: Model Development & Integration (LinkedIn Learning, Apr 2025)
- Apache Spark: Big Data Engineering (LinkedIn Learning, Apr 2025)
- Azure Spark Databricks (LinkedIn Learning, Apr 2025)
- Apache Kafka Essential Training (LinkedIn Learning, Mar 2025)
- SQL (Basic) – HackerRank Certified (Jan 2025)